

Introduction to Macroeconomics (S102016CR)

Spring Semester

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Third Part
Open
Macroeconomics

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Macroeconomics"] --- B["Week 7  
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economy"]
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Week 7
Balance of Payments

Week 8
Exchange rates

Week 9
A macroeconomic
theory of the open
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Week 8:

Exchange rates

- In this second chapter of the third part on open macroeconomics, we will focus on the definition and determination of exchange rates.
- We will start with a discussion of how exchange rates are quoted (direct and indirect quotation)
- We will then distinguish between real and nominal exchange rates
- We will also provide a first model of exchange rate determination in the long run (PPP)
- And then discuss how the exchange rate is determined in the market for foreign currency in the short run

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- a) Real and nominal exchange rates
- b) Determining exchange rates in the long-run: PPP
- c) The exchange rate market and the short-run equilibrium
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a. Real and nominal exchange rates

- International prices influence international transactions.
- The two most critical international prices are the nominal and real exchange rates.
- The *nominal exchange rate* is the rate at which a person can trade one country's currency for another's currency.
- The nominal exchange rate is expressed in two ways:
 - **Direct quotation**: In units of domestic currency (CHF) per unit of foreign currency.
 - **Indirect quotation**: In units of foreign currency per unit of domestic currency (CHF).

Example:

- You can exchange CHF 1.20 Swiss Franc for one Euro.
- The exchange rate is CHF 1.20 per Euro (**direct quotation**).
- The exchange rate is $1/1.20=0.83$ Euros per Swiss Franc (**indirect quotation**)

- *Appreciation* refers to an increase in the value of a currency as measured by the amount of foreign currency it can buy.
- *Depreciation* refers to a decrease in the value of a currency as measured by the amount of foreign currency it can buy.
- **Careful**, because with direct quotation (generally used) when the domestic currency appreciates the value of the nominal exchange rates declines.....
- When the exchange rate of the Euro in terms of CHF goes from CHF 1.20 per Euro to CHF 1, the exchange rate in direct quotation of the CHF with respect to the Euro declines, but the CHF becomes more valuable in terms of Euro. We have an appreciation of the CHF in terms of Euro, and vice-versa a depreciation of the Euro in terms of CHF.

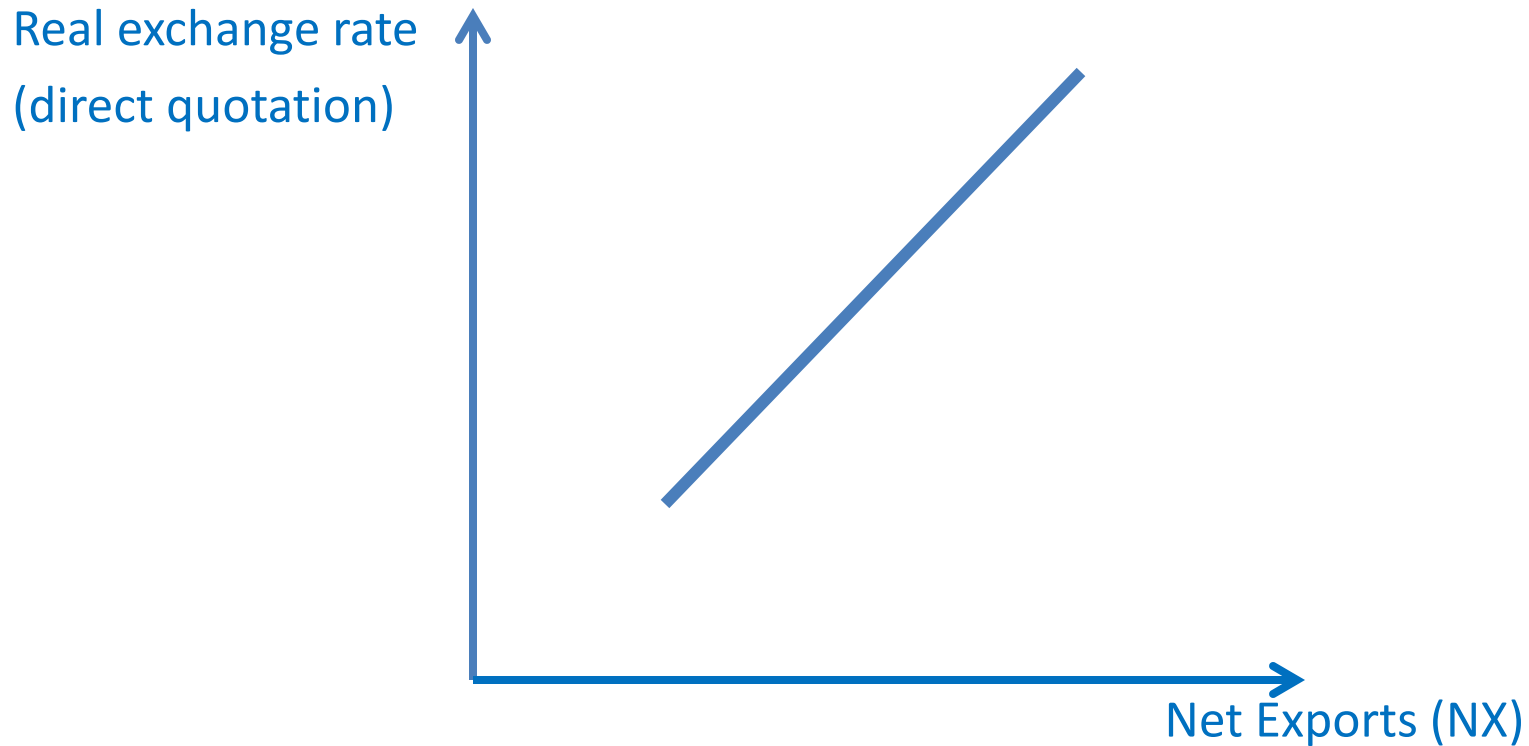
- The *real exchange rate* is the rate at which a person can trade one country's goods and services for another's goods and services.
- The real exchange rate compares the prices of domestic goods and foreign goods in the domestic economy:
 - If a kilo of Swiss cheese is twice as expensive as a kilo of English cheese, the real exchange rate in terms of kilos of cheese is 1/2 a kilo of Swiss cheese per kilo of English cheese.
 - At the macroeconomic level, the real exchange rate is calculated over all foreign and domestic goods using price indices:

$$\text{real exchange rate} = \frac{\text{Nominal exchange rate} \times \text{Foreign prices}}{\text{Domestic prices}}$$

- This implies that for a given nominal exchange rate (like when the Euro was fixed at 1.20 CHF), more rapid inflation abroad than at home leads to an increase in the real exchange rate (real depreciation of the domestic currency), as foreign goods become relatively more expensive than domestic goods (one unit of domestic goods purchases fewer units of foreign goods).

- The real exchange rate is a key determinant of how much a country exports and imports.
- A depreciation (increase in the direct quotation) of the Swiss Franc means that Swiss goods have become cheaper than foreign goods.
- This encourages consumers at home and abroad to buy more Swiss goods and fewer goods from other countries.
- As a result, Swiss exports rise, and Swiss imports fall, and both of these changes raise Swiss net exports.
- Conversely, an appreciation of the Swiss Franc means that Swiss goods have become more expensive than foreign goods, so Swiss net exports fall.

This implies:



Remember: when the real exchange rate increases (in direct quotation), this means that domestic goods have become relatively cheaper, and therefore net exports increase

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b. Determining exchange rates: PPP

- The *purchasing power parity theory* is the most straightforward and popular theory explaining the variation of currency exchange rates in the long run.
- It is based on the idea that 1 unit of domestic currency can buy the same quantity of goods and services in all countries.
- It is based on *the law of one price*, according to which a good must sell for the same price in all locations.
- If the law of one price were not valid, unexploited profit opportunities exist.
- The process of taking advantage of price differences in different markets is called *arbitrage*. If arbitrage occurs, prices converge across markets.
- According to the purchasing power parity theory (PPP), exchange rates adjust to ensure that currencies have the same purchasing power in all countries

- More formally, the law of one price tells us that:

$$P_i = e \times P_i^*$$

- PPP implies:

$$P = e \times P^*$$

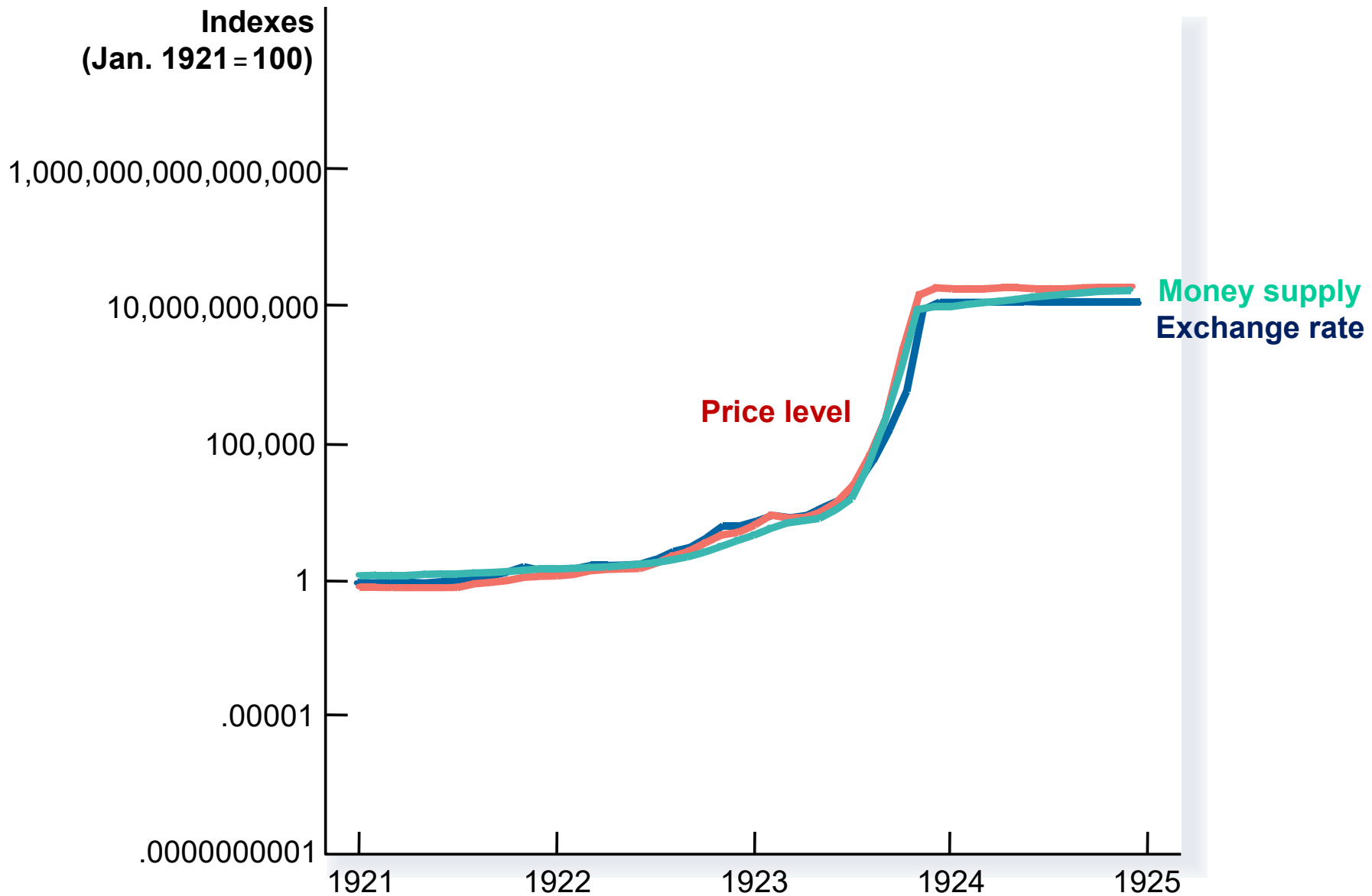
- Solve for e:

$$e = P/P^*$$

- And prices at home and abroad will get determined by the money supply in the long run as we saw with the quantitative theory of money ($MV=PQ$)
- Thus, PPP + quantitative theory of money implies:

$$e = P/P^* = f(MS/MS^*)$$

Money, prices, and the exchange rate during Germany's hyperinflation



- But apart from this example of massive hyperinflation in Germany, casual observation suggests that PPP does not hold. Indeed, goods' prices are not the same in different countries. Why?

Limits to purchasing power parity

- Many goods and services are not easily traded or shipped from one country to another (haircuts in Paris versus haircuts in New York). Thus, arbitrage would be too limited to eliminate price differences across locations.
- Tradable goods are not always perfect substitutes when produced in different countries (Italian versus German cars). There is no opportunity for arbitrage here because the price difference reflects the different value consumers in different countries give to different products.
- Trade policy and trade costs will not allow the law of one price to hold...

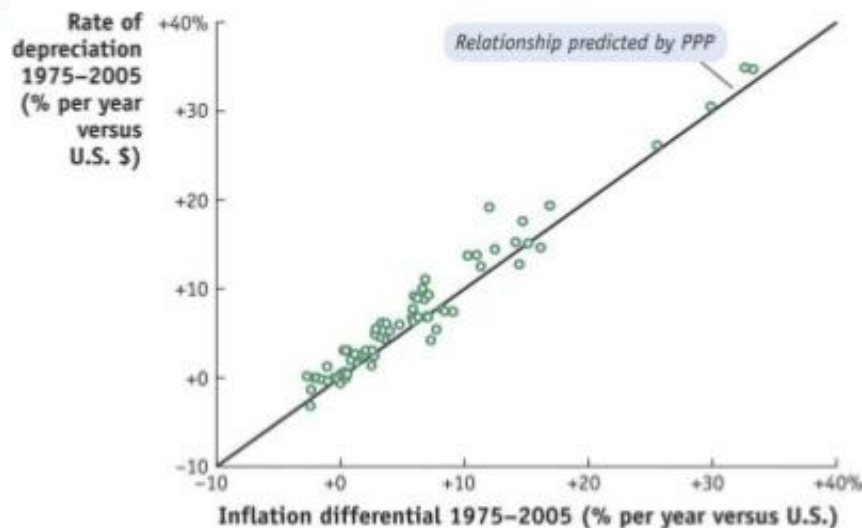
- However, there is more evidence for **relative PPP**, which predicts changes in the nominal exchange rate rather than absolute levels (absolute price levels need not be equal, but price changes must be consistent with PPP in the long run).
- Start from $e = P/P^*$, and take percentage differences to obtain:

$$\% e = \% P - \% P^* \rightarrow \% e = \pi - \pi^*$$

Evidence for PPP in the Long Run

Inflation Differentials and the Exchange Rate, 1975-2005

This scatterplot shows the relationship between the rate of exchange rate depreciation against the U.S. dollar and the inflation differential against the United States over 30 years, for a sample of 82 countries. The correlation between the two variables is strong and bears a close resemblance to the prediction of PPP that all data points would appear on the 45-degree line.



Real exchange rate when the nominal exchange rate is determined by PPP

- An interesting implication of PPP is that the real exchange rate is constant and equal to 1. The relative price of the foreign bundle with respect to the domestic bundle cannot move if PPP applies:

$$\text{real exchange rate} = e \times P^* / P$$

- If PPP holds, then $e = P/P^*$. Substitute it into the above to obtain:

$$\text{real exchange rate} = (P/P^*) \times P^* / P = 1$$

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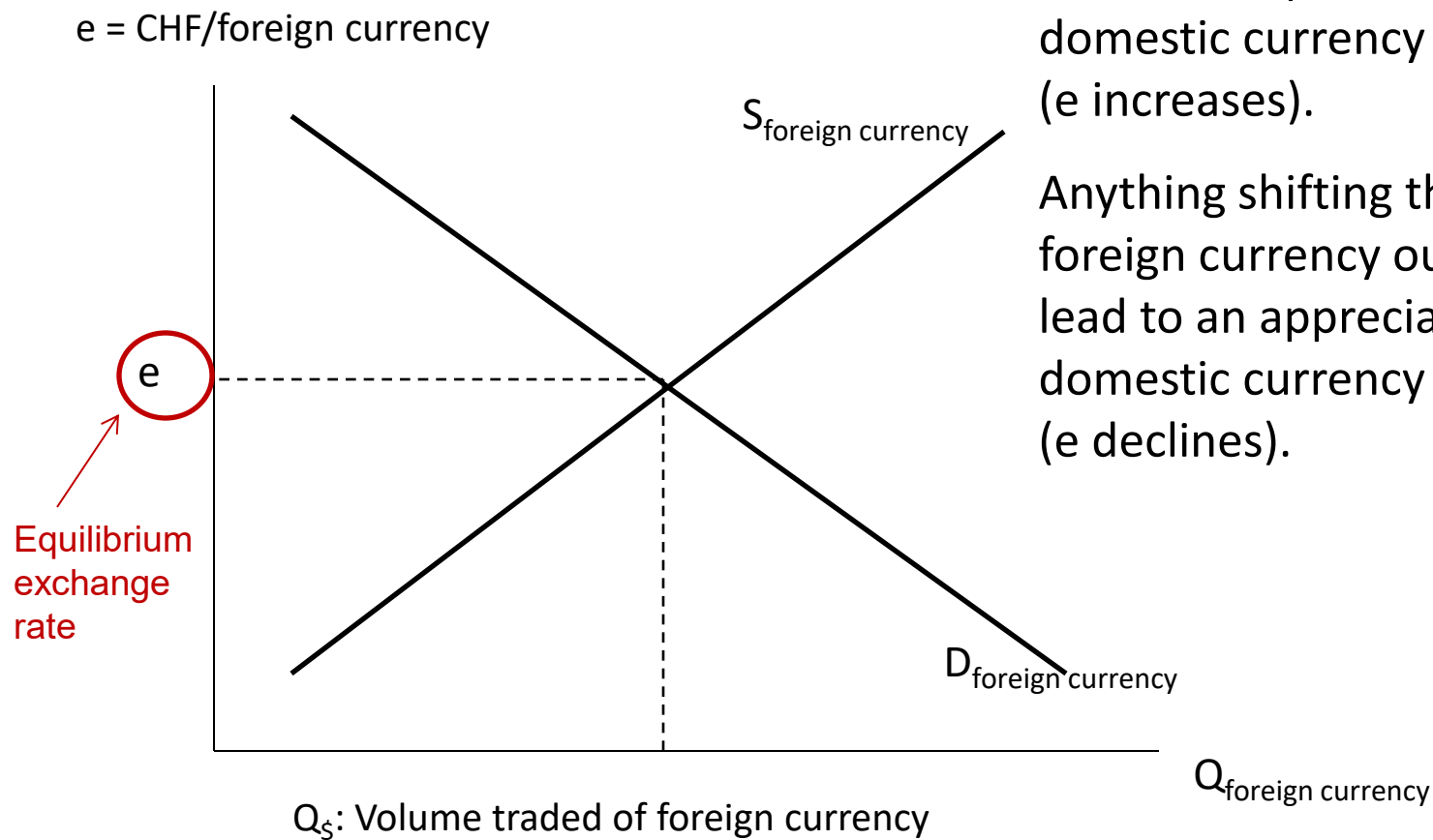
c. The exchange rate market and the short-run equilibrium

- The PPP gives us a determination of exchange rates (the price of foreign currency) in the long run after arbitrage has taken place
- But we also need to understand how exchange rates get determined in the short run. Like in all other markets, it will be determined by the matching between the demand and supply of foreign currency.
- **Supply** is influenced by:
 - Exports of goods&services
 - Factor income abroad
 - Capital inflows
 - Foreign currency sales by CB
 - Etc....
- **Demand** is influenced by:
 - Imports of goods&services
 - Foreign factor income at home
 - Capital outflows
 - Foreign currency purchases by CB
 - Etc....

The market for foreign currency

Anything shifting the demand for foreign currency outwards will lead to a depreciation of the domestic currency in equilibrium (e increases).

Anything shifting the supply of foreign currency outwards will lead to an appreciation of the domestic currency in equilibrium (e declines).



Floating versus fixed exchange rate regimes

- Under a **floating exchange rate regime**, the CB lets the market determine the value of the foreign currency.
- Under a floating/flexible exchange rate regime, if exports increase because of a positive shock to income in the rest of the world, the supply of foreign currency increases, and the exchange rate declines (the domestic currency appreciates) as there is more supply of foreign currency
- Under a **fixed exchange rate regime**, the CB constantly intervenes in the market to keep the price of foreign currency fixed (a recent example is the SNB interventions to keep the Euro at 1.20 CHF between 2011 and 2015, but also the gold standard until 1973, or the EMS before the Euro or the Argentinean 1 peso = 1 USD in the 1990s).
- Under a fixed exchange rate regime, the CB must constantly intervene to absorb a foreign currency's excess supply or demand in the market.

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e. Summary

- The nominal exchange rate is the relative price of a currency in terms of another country's currency.
- In direct quotation (used in this class), it is the amount of domestic currency we need to buy a unit of foreign currency
- In indirect quotation (used in the French class), it is the amount of foreign currency we need to buy a unit of the domestic currency.
- The real exchange rate is the price of goods and services in one country relative to the price of goods and services in another country.
- Because we use direct quotation, we define it in this class as the price of foreign goods and services in terms of domestic goods.
- When the nominal exchange rate changes so that each Swiss franc buys more foreign currency, the Swiss franc is said to appreciate or strengthen (in direct quotation, the exchange rate declines).
- When the nominal exchange rate changes so that each Swiss franc buys less foreign currency, the euro is said to depreciate or weaken (in direct quotation, the exchange rate increases).

- According to the purchasing power parity theory, a unit of currency should buy the same quantity of goods in all countries.
- The nominal exchange rate between the currencies of two countries should reflect the countries' price levels in those countries.
- The theory of relative purchasing power parity is more likely to be verified in the data. It suggests that changes in the nominal exchange rate are equal to the inflation differential between the two countries.
- In the short-run, the exchange rate (the price of foreign currency) gets determined in the market for foreign currency
- Imports, capital outflows, etc., influence demand for foreign currency.
- Supply of foreign currency is influenced by exports, capital inflows, etc....
- With a floating exchange rate regime, the CB does not intervene in the exchange rate market
- With a fixed exchange rate regime, the CB intervenes in the exchange rate market by either buying or selling foreign currency, monetary policy, or capital or exchange rate controls.